# Trung Pham

Principal Research Scientist · NVIDIA — Cosmos Lab · Santa Clara, CA
trungp@nvidia.com · trung.ptt@gmail.com
trungtpham.github.io · Scholar · GitHub · LinkedIn

Principal Research Scientist at **NVIDIA Cosmos Lab**, building **world foundation models for Physical AI**. Track record across **research**, **product delivery**, and **technical leadership**. Ph.D. in Computer Vision; **Google PhD Fellow** and **Heidelberg Laureate Forum** delegate.

## EXPERIENCE

**Principal Research Scientist** — NVIDIA, Cosmos Lab — 2026 – Present

World foundation models for Physical AI. Co-author on **Cosmos World Model**.

**Principal Engineer** — NVIDIA, Autonomous Vehicles — 2024 – 2026

Led BEV perception and online mapping. Designed and shipped **BEV camera perception**, **online mapping**, and **path perception** into NVIDIA's autonomous driving stack. First author of **NVAutoNet** (WACV 2024).

**Senior Computer Vision Scientist** — NVIDIA, Autonomous Vehicles — 2018 – 2024

Designed and shipped core perception models (lanes, roads, intersections, objects, lights, signs) for NVIDIA's production AV stack.

**Postdoctoral Research Fellow** — Australian Centre for Robotic Vision, University of Adelaide — 2014 – 2018

Research on 3D scene understanding and instance segmentation. Core team member on the **1st-place** entry at the **2017 Amazon Robotics Challenge**.

**Software Engineering Intern** — Google — Aug – Oct 2013

Efficient placement algorithms for a global storage system.

## EDUCATION

**Ph.D., Computer Vision** — The University of Adelaide — 2010 – 2014

Computer Vision, Machine Learning. **Google PhD Fellow. Dean's Commendation for Doctoral Thesis Excellence.**

**M.Eng., Computer Engineering** — Chonnam National University — 2008 – 2010

Brain Korea 21 Scholar.

**B.Sc., Mathematics and Computer Science** — Vietnam National University, HCMC

## HONORS

**1st place**, Amazon Robotics Challenge (ACRV team) — 2017

Dean's Commendation for Doctoral Thesis Excellence, Adelaide — 2014

Invited delegate, **Heidelberg Laureate Forum** 2013

**Google PhD Fellowship** in Computer Vision 2012

Adelaide Scholarships International 2010 – 2014

Brain Korea 21 Scholarship 2008 – 2009

## SKILLS

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|---|---|
| EXPERTISE | World foundation models · VLM · VLA · video generation & simulation · radar simulation · BEV perception · online mapping · end-to-end autonomous driving · 3D scene understanding · semantic segmentation · object detection · motion estimation · robust estimation. |
| STACK | Python · PyTorch · C++ · SLURM · large-scale distributed training. |
| LEADERSHIP | Technical lead · mentoring scientists and PhD interns · cross-functional teams. |

## SELECTED PUBLICATIONS

Cosmos 3: Omnimodal World Models for Physical AI 2026

NVIDIA. *NVIDIA Technical Report*.

World Simulation with Video Foundation Models for Physical AI 2025

NVIDIA. *arXiv:2511.00062*.

NVAutoNet: Fast and Accurate 360° 3D Visual Perception for Self-Driving 2024

**T. Pham**, M. Maghoumi, et al. *WACV 2024*.

Bayesian Semantic Instance Segmentation in Open Set World 2018

**T. Pham**, V. B. G. Kumar, T.-T. Do, G. Carneiro, I. Reid. *ECCV 2018*.

Cartman: The Low-Cost Cartesian Manipulator that Won the Amazon Robotics Challenge 2018

D. Morrison, ..., **T. Pham**, et al. *ICRA 2018*.

A Bayesian Data Augmentation Approach for Learning Deep Models 2017

T. Tran, **T. Pham**, G. Carneiro, L. Palmer, I. Reid. *NeurIPS 2017*.

Meaningful Maps — Object-Oriented Semantic Mapping 2017

N. Sünderhauf, **T. Pham**, Y. Latif, M. Milford, I. Reid. *IROS 2017*.

Efficient Point Process Inference for Large-Scale Object Detection 2016

**T. Pham**, H. Rezatofighi, T.-J. Chin, I. Reid. *CVPR 2016*.

The Random Cluster Model for Robust Geometric Fitting 2013

**T. Pham**, T.-J. Chin, J. Yu, D. Suter. *IEEE TPAMI 2013*.

## SERVICE

Reviewer for top-tier computer vision, machine learning, and robotics venues — **CVPR**, **ICCV**, **ECCV**, **NeurIPS**, **ICRA**, **IROS**, **IEEE TPAMI**, and **IEEE TIP**.